

Assessment of the impact of dissemination of genetically improved Abbassa Nile tilapia strain (GIANT-G9) versus commercial strains in some Egyptian governorates

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Abstract

WorldFish initiated a selective breeding program in Abbassa—Egypt to develop and produce the genetically improved Nile tilapia strain known as “Genetically Improved Abbassa Nile tilapia (GIANT)”, adopting the same technology used for the Genetically Improved Farmed Tilapia (GIFT), in Asia. WorldFish provided the Ninth Generation (G9) broodstock of the GIANT to 11 Broodstock Multiplication Centers (BMC’s) in five governorates; these centres then disseminated improved mixed-sex fry to 160 tilapia hatcheries which supplied all-male fry to 1,500 fish farms in 2017. We present results of an impact assessment of the G9 of GIANT in 83 of these fish farms in four Egyptian governorates (Kafr El-Sheikh, Fayoum, Behera and Alexandria). Results indicate that the use of GIANT in all governorates achieved significantly higher (18.8%) fish yields (12.3%–26.4%) and 15.7% lower FCR (feed conversion ratio), reflecting a positive impact on their net profits compared with those who used commercial strains. Overall, fish farmers who stocked GIANT had significantly higher average fish sales (5,567 US\$/ha) than those who stocked non-improved commercial strains (5,192 US\$/ha) in all governorates. However, when comparing within governorates, the difference was significant only in Fayoum and Alexandria noting a mixed performance of the GIANT strain. The mixed performance indicates that some hatcheries may have crossed the different strains, which lowered the performance differences. To get an objective analysis of the performance of the GIANT, it is recommended that similar assessment be undertaken with new generations of the strain supported by on-farm performance trials.

KEYWORDS

comparison, dissemination, economic, genetic improvement, growth, Tilapia

1 | INTRODUCTION

World tilapia aquaculture production grew 12 percentage annually, from less than a half million tons in the early 1990s to about 5.4 million tons in 2016 (FAO, 2018). Africa accounted for 20 percentage of the growth with Egypt contributing the bulk (80%) of this production. Egyptian fish production in 2016 was about 1.7 million tons, 1.1 million tons (or 65%) of which were tilapia (GAFRD, 2016).

The growing demand for fish, which is driven by population and economic growth in Africa, is an indication of great potential for the development of fish farming in Africa and in sub-Saharan Africa in particular (FAO, 2017). A rapid expansion of aquaculture production is seen as the answer for filling the growing gap between supply and demand for fish across the continent (Beveridge et al., 2013).

Several genetic improvement programs have been implemented and developed for Nile tilapia, and other strains such as Genetically

Improved Farmed Tilapia (GIFT; Eknath & Acosta, 1998; Eknath et al., 1993), FaST (Bolivar, 1998), and GET-EXCEL are also available (Tayamen, 2004). All of these programs were however, implemented in Asia. In Africa and the Middle East, where Nile tilapia is native, farmed tilapia has, until recently, been based on non-genetically improved strains (Khaw et al., 2009). Dey, Eknath, et al. (2000) reported that the introduction of GIFT strain into five Asian countries (Bangladesh, China, the Philippines, Thailand and Vietnam) has led to increase in fish yield due to higher average weight at harvest. The bigger tilapia market size led to higher sale prices and farm revenue (Dey, 2000).

A selective breeding program was initiated in 2001 at Abbassa—Egypt to develop and produce a genetically improved Nile tilapia which was to be referred to as “Genetically Improved Abbassa Nile Tilapia (GIANT) strain”, using the same technology that produced GIFT (Rezk et al., 2009). This program was designed to suit the needs of Egyptian and African aquaculture conditions. The GIANT strain was selected for growth and high survival rates since most tilapia farmers consider fast growth from seed to harvest size to be the most important performance trait, along with high survival rate (Ibrahim, Abou Zaid, Khaw, El-Naggar, & Ponzoni, 2013).

On-station comparison of generation 8th and 9th of GIANT strain with commercial strain at Abbassa showed that GIANT strain performance was 28% faster than the commercial strain. Based on this result, the GIANT selection line was released to the local aquaculture sector (Ibrahim et al., 2013). A dissemination strategy was adopted to disseminate the G9 of GIANT strain countrywide (Dickson, Nasr-Allah, Kenawy, & Kruijsen, 2016a). Almost 38% of the total Egyptian fish farms received and used GIANT strain by 2017 according to the midterm review of the Sustainable Transformation of Egypt's Aquaculture Market System (STREAMS) project.

The aim of the present study was to assess the impact of the ninth generation of GIANT strain at farm level after its dissemination to fish farms in Egypt.

2 | METHODOLOGY

2.1 | Dissemination of improved Abbassa Nile tilapia strain

In June 2012 WorldFish distributed mixed sex fingerlings (average weight of 1 g) of the improved Nile tilapia (*Oreochromis niloticus*) strain for free to five broodstock multiplication centres (BMCs) in four governorates, namely Kafr el Sheikh, Behera, Fayoum and Sharkia, which have significant aquaculture industries. Later six BMCs were added including one in El Minya, (Upper Egypt). Broodstock multiplication centres (BMCs) were selected based on the geographical distribution, experience and their commitment to providing data on seed dissemination. The BMCs reared the fingerlings to maturity in preparation for spawning in the following season. In May/June 2013, the BMCs multiplied the improved strain and disseminated mixed-sex fry to 160 commercial hatcheries (40% of hatcheries according to GAFRD, 2012). In 2014, commercial hatcheries supplied all-male

fry and fingerlings to fish farmers across the target governorates. Around 459 fish farms were stocked with 109 million fry of the improved strain in 2014.

This was the first time that the improved strain was ever released to fish farmers in Egypt. The 9th generation of the genetically improved strain was derived from 114 half-sib families.

2.2 | Questionnaire preparation and data collection

Eighty-three fish farms in four governorates (Kaf El-Sheikh, Fayoum, Behera and Alexandria) from the WorldFish farmer database were selected based on evidence of having kept pure strains without mixing with other strains. These farms consisted of 303 fishponds in a total area of 857 Feddan (360 hectare).

A detailed questionnaire was developed to enable the generation of data on pond production parameters and some financial parameters. Growth parameters studied include stocking density, average initial weight (AIW), gross yield, net yield, daily weight gain, grow-out period, pond area and Apparent Food Conversion Ratio (AFCR). The economic parameters are, gross revenue (fish sales), average sale price, operational cost, feed cost, return above operational cost, break-even price and safety margin. Data were classified into two groups: ponds stocked with the improved Abbassa strain (GIANT); and ponds stocked with the local commercial strains (Com). Table 1 shows the two groups across the four governorates. Data were initially checked for validity and where necessary the data collectors had to ask fish farmers for further clarifying information. In case of ponds with declared mixed strains (GIANT + Com) in same pond, the data were excluded.

2.3 | Data analysis

Growth and economic parameters were analysed using General Liner Model (GLM) procedure in SAS. The least squares means were also computed for growth and economic parameters, and differences were considered significant at $p \leq .05$. Since the total fish yield and net fish yield were not normally distributed, we transformed the data to square root values for data normalization. The average initial weight (AIW) differed significantly from pond to pond (Table 2) indicating that there was a confounding effect by the AIW, we used the AIW as a covariate in the analysis of the growth parameters. All growth parameters were calculated as ton per hectare (ton/ha), and

TABLE 1 Number of ponds used in this study stocked with Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains by governorates

Governorate	GIANT	Commercial	Total
Kafr El-Sheikh	15	67	82
Fayoum	67	29	96
Behera	12	63	75
Alexandria	26	24	50
Total	120	183	303

TABLE 2 Least squares means (LSM) $\pm$ SE of average initial weight (AIW), stocking density, total yield, net yield (back transformed to actual units from the square root scale), daily weight gain, grow-out period, pond size and feed conversion ratio (FCR) for Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains in four Egyptian governorates

Variable	Kafr El Sheikh		Fayoum		Behera		Alexandria	
	GIANT (15)	Commercial (67)	GIANT (67)	Commercial (29)	GIANT (12)	Commercial (63)	GIANT (26)	Commercial (24)
AIW (g)	0.11 $\pm$ 0.21	4.72 $\pm$ 0.99	0.5 $\pm$ 0.01	0.63 $\pm$ 0.03	0.11 $\pm$ 0.37	20.42 $\pm$ 1.60	2.69 $\pm$ 0.40	3.00 $\pm$ 0.41
<i>p</i> -value	.0497		.0001		.0001		.5946	
Stock. density (1,000 fish/ha)	37.9 $\pm$ 3.4	31.6 $\pm$ 1.6	30.9 $\pm$ 0.5	27.9 $\pm$ 0.7	47.4 $\pm$ 5.9	49.6 $\pm$ 2.4	59.8 $\pm$ 2.9	55.1 $\pm$ 3.0
<i>p</i> -value	.0961		.0008		.7408		.2700	
Total yield (ton/ha)	7.35 $\pm$ 0.1	6.53 $\pm$ 0.3	5.74 $\pm$ 0.1	4.65 $\pm$ 0.2	11.30 $\pm$ 0.3	9.36 $\pm$ 0.5	7.15 $\pm$ 0.8	5.70 $\pm$ 0.9
<i>p</i> -value	.1263		<.0001		.1237		.0308	
Net yield (ton/ha)	7.24 $\pm$ 0.1	6.42 $\pm$ 0.2	5.73 $\pm$ 0.1	4.63 $\pm$ 0.2	10.05 $\pm$ 0.3	8.95 $\pm$ 0.4	6.98 $\pm$ 0.8	5.52 $\pm$ 0.9
<i>p</i> -value	.1266		<.0001		.3273		.0296	
Daily Wt. gain (g/day/ha)	22.9 $\pm$ 1.8	24.5 $\pm$ 0.9	24.9 $\pm$ 0.5	20.3 $\pm$ 0.8	33.1 $\pm$ 4.5	39.9 $\pm$ 1.9	21.0 $\pm$ 1.4	16.7 $\pm$ 1.4
<i>p</i> -value	.4533		<.0001		.1832		.0326	
Grow-out period (days)	332.3 $\pm$ 22.9	286.3 $\pm$ 10.8	234.06 $\pm$ 4.6	234.1 $\pm$ 6.6	316.2 $\pm$ 15.3	250.6 $\pm$ 6.2	342.5 $\pm$ 14.1	371.5 $\pm$ 14.7
<i>p</i> -value	.0751		.9962		.0003		.1621	
Pond size (ha)	1.76 $\pm$ 0.37	1.29 $\pm$ 0.18	0.72 $\pm$ 0.02	0.76 $\pm$ 0.02	2.27 $\pm$ 0.51	1.55 $\pm$ 0.22	1.13 $\pm$ 0.06	1.18 $\pm$ 0.07
<i>p</i> -value	.2539		.2547		.2027		.6202	
AFCR	1.67 $\pm$ 0.18	1.88 $\pm$ 0.08	1.43 $\pm$ 0.02	1.58 $\pm$ 0.04	1.48 $\pm$ 0.11	1.92 $\pm$ 0.05	1.22 $\pm$ 0.04	1.50 $\pm$ 0.4
<i>p</i> -value	.2907		.0012		.0009		<.0001	

Note: *p*-value indicated in bold are those significant at the level of less than or equal .05.

all economic parameters were calculated as US dollar per hectare (\$/ha) or US dollar per kg (\$/kg). Gross revenue was the summation of all sales value of tilapia classes, total operational cost was the summation of all operational costs incurred during the breeding season.

Return above operational costs, break-even price and safety margin were calculated according to Jolly and Clonts (1993) as follows:

Return above operational costs = Gross revenue – Total operational cost.

$$\text{Break-even price} = \frac{\text{Total operational cost}(\$)}{\text{Total yield (kg)}}$$

$$\text{Safety margin percent} = \frac{\text{Sales price}(\$) - \text{Break-even price}}{\text{Sales price}} \times 100.$$

All statistical routines were carried out using SAS ver. 9.4 (SAS Institute, 2017) according to Steele and Torrie (1980).

3 | RESULTS

3.1 | Fish growth and yield

Fish farmers stocked GIANT strain at higher stocking density (4.3 fish/m²) than that of commercial strain (3.8 fish/m²) in all governorates except Behera, where ponds were stocked with almost the same average stocking density (4.8 fish/m²) for both (Table 2).

The total fish yield per ton was between 4.65 and 11.30 ton/ha with the highest recorded yield for the improved strain in Behera. The trend was similar for net fish yield which was between 4.63 and 10.05 ton/ha. However, the daily weight gain was highest for the non-improved commercial strain at 39.9 kg day⁻¹ ha⁻¹ and lowest (16.7 kg day⁻¹ ha⁻¹) for the same strain in Alexandria. The growth performance parameters for both GIANT and non-improved strains were highest in Behera compared with all other governorates indicating better overall management. Total fish yield and net fish yield were significantly greater ($p < .05$) for improved strain than for commercial strain in Fayoum and Alexandria but not significant ($p > .05$) in Kafr El-Sheikh and Behera (Table 2 and Figure 1). Daily weight gain (DWG) was significantly higher in the GIANT strain than in the commercial strain in Fayoum and Alexandria, but the opposite was true in Kafr El-Sheikh and Behera governorates (Table 2).

GIANT ponds demonstrated efficient feed conversion ratio (Table 2). The results indicate a significantly lower overall APCR ($p < .05$) in GIANT ponds (1.45) compared with commercial ponds (1.72; Figure 2). Apart from Kafr El Sheikh, APCR was significantly lower ($p < .01$) in GIANT strain than the commercial strain (Table 2).

3.2 | Economic impact

Partial economic analyses of the performance of the GIANT strain against commercial strain are presented in Tables 3, 4 and 5). Fish farmers in all governorates who stocked GIANT strain had higher average gross revenue (fish sales) and lower total operational costs than those who stocked commercial strains (Table 5). The overall average revenue was significantly higher ($p < .05$) with GIANT strain than commercial strain (5,567 and 5,192 \$/ha respectively). On the other hand, the overall average operational costs were lower although not significant between GIANT and commercial strains (3,413 and 3,742 \$/ha respectively). Overall, the average return above operational costs was significantly higher ($p < .01$) in GIANT than commercial strains (2,154 and 1,450 \$/ha, respectively) as presented in Table 5 and illustrated in Figure 3.

Fish farmers using commercial strain spent significantly more on fish feeds (3,075.1 $\pm$ 96.2 \$/ha) than those who used GIANT strain (2,638.5 $\pm$ 106.6 \$/ha). The contribution of feed to the total operational cost in the fish farm was significantly higher for fish farmers using commercial strain (82.2%) than those using the GIANT strain (77.3%). Thus, although feed was the highest cost item for both strains (Table 4), the lower APCR for the GIANT strain indicated it to be the more efficient strain in all governorates.

Labour costs represented 9.6% and 5.7% of the total costs for GIANT and commercial strains, respectively, making it the second largest cost item (Table 4). Labour costs were significantly higher for GIANT strain (327.5 \$/ha) than commercial (213.7 \$/ha). At 167.7 \$/ha (4.9% of total operational costs) and 156.1 \$/ha (4.2%) for GIANT and commercial strain, respectively, seed was third largest cost item.

As shown in Table 5, average sale price of GIANT strain was 0.87 $\pm$ 0.02 \$/kg which was significantly higher than commercial strain (0.78 $\pm$ 0.02 \$/kg). The cost of feed for producing a kg of fish yield and break-even price were significantly lower with GIANT

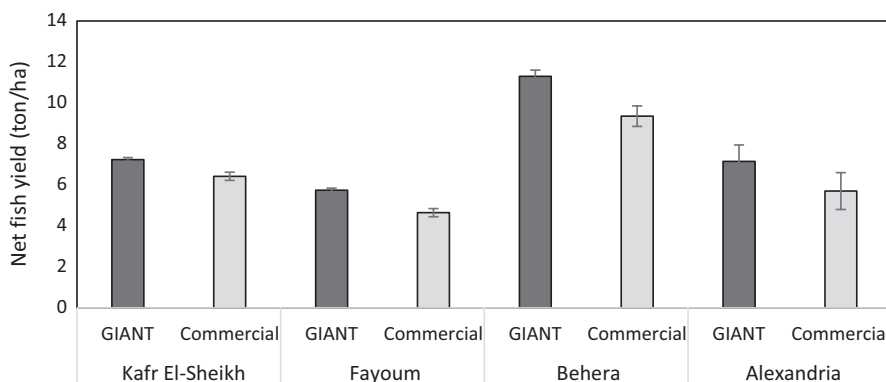
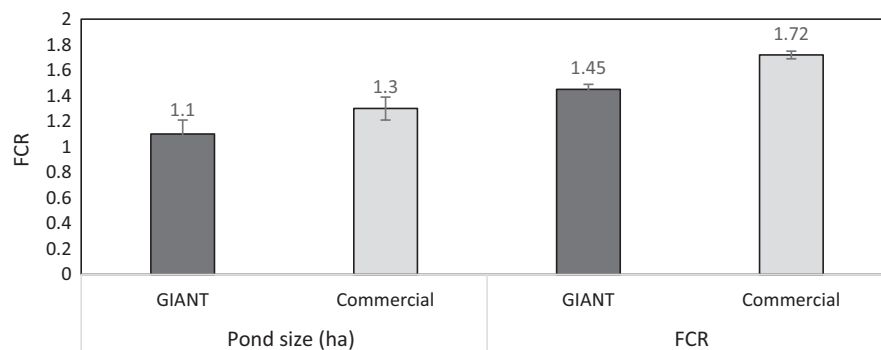


FIGURE 1 Net fish yield (ton/ha) for Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains in different Egyptian governorates

FIGURE 2 Feed conversion ratio for Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains in all studied governorates



(0.43 and 0.52 \$/kg, respectively) than that with commercial strain (0.48 and 0.56 \$/kg respectively).

Safety margin percentage and operational profit as percentages of total costs were significantly higher with GIANT (35.4%) than commercial (23.8%) indicating that GIANT generated more profit with higher safety margins for fish farmers compared with commercial strains.

4 | DISCUSSION

Results of the present study indicate that fish farmers who used GIANT in the governorates, achieved higher fish yield and better feed conversion ratio, reflecting a positive impact on their net profits compared with the fish farmers who used other commercial unimproved strains. In an earlier study, the seventh and eighth generations of GIANT had shown a superiority of 28% in growth rate over the commercial strain in on-station trials (Ibrahim et al., 2013). The present study confirmed the superiority of the GIANT at farm level although with marked differences between governorates. These differences may have been due to the differences in management systems that fish farmers applied in different governorates. This may suggest an existence of genotype by environmental interaction contrary to Khaw et al. (2009) who working with the same selected strain in two systems differing in level of input, concluded that there was no evidence for genotype by environmental interaction. In another study, Charo-Karisa et al. (2006) using genetically similar fish showed that pond effects could significantly affect fish performance. This suggests that Genotype by Environment Interactions may exist if the testing environments differ substantially in level of inputs.

In a similar study in five different Asian countries, Dey, Eknath, et al. (2000) reported yield differences ranging from 8.7% to 77.8% among pond farmers using GIFT. They also reported that the introduction of GIFT led to decrease in variable costs ranging from 0.7% to 14.3% in some Asian countries, while in Thailand variable costs increased by 0.5%. Dey et al. (2013) also studied the impact of the genetically improved carp in different Asian countries. They found differences in performance of the improved strains among countries reporting that the adoption of improved carp strain would increase production of silver barb by 31%–48% in Bangladesh and 19%–31%

in Thailand; rohu by 63%–73% in India, and common carp by 45%–71% in Vietnam, and 3%–11% in China.

We found no significant differences in the stocking densities within all governorates except in Fayoum, where GIANT was stocked at higher average densities (30,900 fish/ha) and commercial strain was stocked at lower average densities (27,900 fish/ha). This was mainly because fish farmers stocked GIANT at lower average initial weight (0.5 g) than the commercial strain (0.63 g). The lower initial stocking size for the GIANT (15.43 kg/ha) compared with commercial strain (17.51 kg/ha) could affect the final biomass of the GIANT negatively at the end of the growth period. Despite the lower initial biomass of GIANT against the commercial strains, net fish yield was higher in the case of GIANT strain than commercial strain in all governorates with an overall average of about 18.8% (Table 2). Dey, Eknath, et al. (2000) reported that stocking rate of GIFT strain in fish ponds in Asian countries were similar except for Philippines, in which stocking rate was higher for non-GIFT ponds. The authors reported that stocking size vary from country to country. The highest stocking size reported in China 56.8g the lowest was in Philippines 0.76g.

Zhang, Zhu, and Zhou (1987) indicates that smaller ponds are more manageable and productive per unit area than bigger ones. The lower degree of superiority of GIANT strain over the commercial in Kafr El-Sheikh and Behera governorates may be due to the larger pond sizes used in these two governorates for GIANT (1.76 and 2.27 ha) compared with commercial (1.29 and 1.55 ha).

One explanation for the difference in the grow-out period in Behera, between GIANT, which lasted longer - an average of 316.2 days, and commercial strain, which lasted for an average of 250.6 days, relates to the AIW. Fish farmers in Behera stocked GIANT fry at lower AIW, 0.11 g, while they stocked the commercial strain at higher AIW, 20.42 g (Table 2). Farmers stocked their commercial strains at fingerling stage, which may have overwintered in some cases from the last year but bought GIANT fry produced that same year from the relevant BMCs thus leading to lower AIW for GIANT in all governorates.

Fish farmers using GIANT strain achieved significantly lower AFCRs than those who used commercial strains with an overall average of about 15.7% (Table 2 and Figure 3) indicating that the improved strain used the artificial feed more efficiently than the other commercial strains. This has resulted to a significant reduction in GIANT strain feeding cost than commercial strains with an overall

TABLE 3 Least squares means (LSM) $\pm$ SE of feed conversion ratio (FCR), gross revenue (fish sales value), operational cost, return above operational and pond size for Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains in four Egyptian governorates

Variable	Kafr El Sheikh		Fayoum		Behera		Alexandria	
	GIANT (15)	Commercial (67)	GIANT (67)	Commercial (29)	GIANT (12)	Commercial (63)	GIANT (26)	Commercial (24)
G. revenue (\$/ha)	6,339 $\pm$ 599	5,379 $\pm$ 2,856	5,638 $\pm$ 142	4,556 $\pm$ 215	8,895 $\pm$ 1,015	7,197 $\pm$ 463	4,297 $\pm$ 270	3,365 $\pm$ 289
<i>p</i> -value	.1516		<.0001		.1369		.0236	
Operational cost (\$/ha)	3,989 $\pm$ 285	3,878 $\pm$ 136	3,135 $\pm$ 39	2,749 $\pm$ 60	6,936 $\pm$ 753	5,363 $\pm$ 343	3,120 $\pm$ 187	2,914 $\pm$ 200
<i>P</i> -value	.7266		<.0001		.0684		.4585	
Return above operational cost (\$/ha)	2,350 $\pm$ 421	1,501 $\pm$ 201	2,503 $\pm$ 125	1,806 $\pm$ 190	1,959 $\pm$ 435	1834 $\pm$ 199	1,177 $\pm$ 136	451 $\pm$ 145
<i>p</i> -value	.0275		.0029		.7950		.0007	
Average sale price (\$/kg)	0.86 $\pm$ 0.05	0.79 $\pm$ 0.02	0.98 $\pm$ 0.01	0.97 $\pm$ 0.02	0.87 $\pm$ 0.07	0.70 $\pm$ 0.03	0.58 $\pm$ 0.00	0.56 $\pm$ 0.01
<i>p</i> -value	.2023		.7232		.0294		.0175	
Feed cost (\$/kg)	0.48 $\pm$ 0.03	0.51 $\pm$ 0.01	0.44 $\pm$ 0.01	0.48 $\pm$ 0.01	0.51 $\pm$ 0.05	0.47 $\pm$ 0.02	0.35 $\pm$ 0.01	0.41 $\pm$ 0.02
<i>p</i> -value	.3248		.0040		.4595		.0007	
Break-even price (\$/kg)	0.55 $\pm$ 0.02	0.59 $\pm$ 0.01	0.56 $\pm$ 0.01	0.60 $\pm$ 0.01	0.66 $\pm$ 0.07	0.56 $\pm$ 0.03	0.44 $\pm$ 0.01	0.48 $\pm$ 0.01
<i>p</i> -value	.1170		.0265		.1702		.0546	
Safety margin \$/kg	31.6 $\pm$ 4.3	21.8 $\pm$ 2.0	42.2 $\pm$ 1.4	37.6 $\pm$ 2.0	23.9 $\pm$ 5.2	19.6 $\pm$ 2.3	23.4 $\pm$ 2.18	14.0 $\pm$ 2.7
<i>p</i> -value	.0401		.0608		.4533		.0096	

Note: *p*-value indicated in bold are those significant at the level of less than or equal .05.

TABLE 4 Comparison of operational costs (LSM \$/ha $\pm$ SE) for farms used Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains

item	GIANT	%	Commercial	%	p value
Feed	2,638.5 $\pm$ 106.6	77.3	3,075.1 $\pm$ 96.2	82.2	.0055
Labor	327.5 $\pm$ 15.9	9.6	213.7 $\pm$ 13.5	5.7	.0003
Seed	167.7 $\pm$ 13.0	4.9	156.1 $\pm$ 11.5	4.2	.4437
Pond preparation	70.4 $\pm$ 6.6	2.1	113.3 $\pm$ 6.0	3.0	<.0001
Fuel	46.7 $\pm$ 10.1	1.4	55.0 $\pm$ 10.3	1.5	.5866
Fertilizers & Vet treatment	63.0 $\pm$ 8.7	1.8	93.3 $\pm$ 8.4	2.4	.0015
Transportation	59.3 $\pm$ 6.9	1.7	29.8 $\pm$ 5.3	0.8	<.0001
Others	40.1 $\pm$ 3.9	1.2	6.0 $\pm$ 1.0	0.2	<.0001
Total	3,413.2 $\pm$ 119.6	100.0	3,742.3 $\pm$ 106.7	100.0	.1504

Note: p-value indicated in bold are those significant at the level of less than or equal .05.

average of about 14% which in turn led to about 5% less of the total costs (Table 4). A similar trend was observed by Dickson, Nasr-Allah, Kenawy, Fathi, et al. (2016b) who recorded a 16.7% lower FCR with GIANT than commercial strains. In the present study large proportions of the operational costs with both strains were spent on fish feed (Tables 3 and 4). Azaza et al., (2009) and El-Sayed, Abdullah, Nasser, Abdel-Wahab, and Yousef (2018) noted similar findings under intensive and semi-intensive aquaculture regimens for Nile tilapia, and stated that, feed constitutes the largest proportion of the overall costs. Dickson, Nasr-Allah, Kenawy, and Kruijssen (2016a) found that the lower feed costs resulted in significantly lower operating costs and feed costs represented 79% of variable costs in control farms. Eltholth, Fornace, Grace, Rushton, & Häslar, 2015

TABLE 5 Financial summary for Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains fish farms (LSM $\pm$ SE)

Financial item	GIANT	Commercial	p-value
Gross revenue (\$/ha)	5,567 $\pm$ 194	5,192 $\pm$ 181	.0411
Operational costs (\$/ha)	3,413 $\pm$ 120	3,742 $\pm$ 107	.1504
Return above operational costs (\$/ha)	2,154 $\pm$ 122	1,450 $\pm$ 108	<.0001
Average sale price (\$/kg)	0.87 $\pm$ 0.02	0.78 $\pm$ 0.02	.0008
Feed cost (\$/kg)	0.43 $\pm$ 0.01	0.48 $\pm$ 0.01	.0001
Break-even price to cover TOC (\$/kg) ^a	0.52 $\pm$ 0.02	0.56 $\pm$ 0.01	.0230
Safety margin (\$/kg)	35.37 $\pm$ 1.46	23.76 $\pm$ 1.30	.0001
Operational profit (% of costs)	63%	39%	–

Note: p-value indicated in bold are those significant at the level of less than or equal .05.

^aBreak-even price to cover variable expenses = Total operating Cost/Yield.

suggested that fish feed availability, quality and prices were the major constraints for tilapia production in Egypt.

We found significantly higher labour cost for farming the GIANT strain than that for commercial strains in the present study. This may be due to the difference of fish yield quantity between both strains, which was higher with GIANT than that with commercial strains. The average total fish yield was 7.9 ton/ha for GIANT and 6.6 ton/ha for commercial strains, so rearing the GIANT strain may have needed more labourers whether full-time, part-time or seasonal for management and harvest. Macfadyen et al. (2012) estimated that for every 100 tonnes of fish produced by pond fish farms, once the product has travelled through the value-chain, it has resulted in almost 14 full equivalent time FTE jobs being created "or needed". The increased labour and transportation costs may have been related to the harvest and the post-harvest handling required for the improved strain.

Seed cost for GIANT and commercial strains was almost the same, however the fry initial weights and consequently their prices at stocking differed. The overall AIW was 7.2 g for commercial and only 0.9 g for GIANT. This was mainly due to the market sale price of GIANT improved fry, which is about 20% higher than the commercial strains when sold at the same sizes. Nasr-Allah, Dickson, Kenawy, Fathi, and El-Naggar (2014) in studying the technical and economic characteristics of different Egyptian Nile tilapia hatchery systems, found that production costs and revenues varied with changes in management systems.

Fish farmers who used GIANT achieved significantly higher average revenue with almost same operational cost compared with those who used the commercial strains. In a survey of farms using GIANT and other strains, Dickson, Nasr-Allah, Kenawy, Fathi, et al. (2016b) found a small increase in overall production of 3.7% with GIANT strain but a significantly higher (68.3%) profit with GIANT over other strains. The main factor contributing to improved fish farm profitability was efficient feed management (Dickson, Nasr-Allah, Kenawy, & Kruijssen, 2016a). Wahab, Kadir, Milstein, and Kunda (2011) studied fish production in polyculture systems and concluded that the increased fish biomass and the reduction in FCR together resulted an increase in total income. These results clearly suggest that increased revenue together with reduction in AFCR were the main factors for

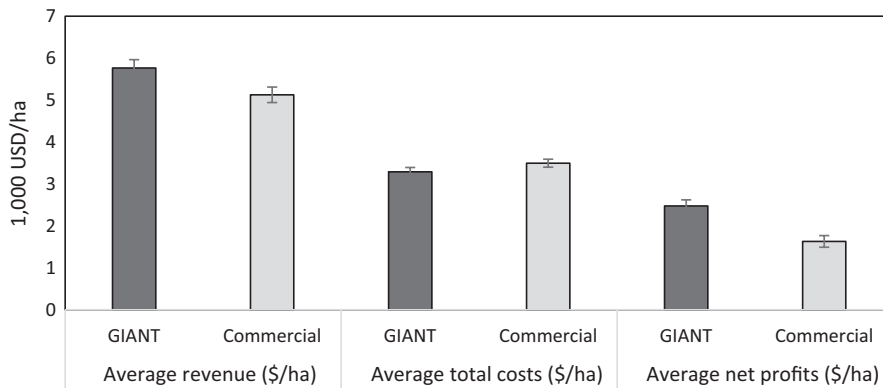


FIGURE 3 Average revenue, operational costs and return above operational costs (1,000 \$/ha) for fish farmers used Genetically Improved Abbassa Nile tilapia (GIANT) and commercial strains

higher average profit about 48.6% for the GIANT strain in the present study.

5 | CONCLUSION

The present study has demonstrated that fish farmers using the GIANT strain achieved higher net fish yield, lower FCRs and higher returns which somehow indicates superiority of the improved strain against the commercial strains. Although the current study was undertaken as a survey and thus was unable to measure subtle and unique attributes of the farms and ponds, there appears to be a strong rationale for the dissemination of superior generations of GIANT to more fish farms. However, this requires further tests given the limitations in the present survey. We recommend real-time data collection methods accompanied by experimental on-farm tests across the country.

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